Waveform-based timing correction for time-of-flight positron emission tomography

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Abstract

Accurate coincidence timing is essential for time-of-flight positron emission tomography. This study investigates machine-learning corrections to leading-edge timing using waveforms from two amplification stages and two readout boards, comparing different estimators, detector-shared formulations, and waveform windows.

Waveform-based correction improves first-stage coincidence timing resolution (CTR) by over 20%, reaching approximately 25% with extended windows. Linear models perform competitively with neural networks, while detector-shared estimators retain most of the achievable correction. Wider windows reveal useful timing information beyond the leading edge. Nevertheless, second-stage signals remain superior, achieving corrected CTRs of approximately 53 ps.

1 Introduction

Positron emission tomography (PET) detects pairs of annihilation photons to reconstruct radiotracer distributions. Time-of-flight (TOF) information uses the difference between photon arrival times to constrain the annihilation position along the line joining the detectors and can improve image signal-to-noise performance. The precision of this information depends on the coincidence timing resolution (CTR), conventionally expressed as the full width at half maximum (FWHM) of the time-difference distribution [1].

A leading-edge discriminator (LED) assigns an arrival time when a pulse crosses a fixed amplitude threshold. Variations in pulse amplitude and rising-edge shape shift the crossing time, producing time walk. Digitized waveforms retain information about these variations, which machine learning (ML) can exploit to estimate event-specific timing corrections.

Waveform-based approaches include direct TOF estimation and correction of conventional timing measurements. Berg and Cherry demonstrated direct TOF estimation from detector waveform pairs using a convolutional neural network (CNN), improving CTR by 20% relative to LED, from 231 to 185 ps [2]. Onishi et al. combined LED with a CNN trained using waveforms acquired at a single source position, improving timing resolution while avoiding the position-dependent bias associated with direct TOF estimation [3]. Feng et al. explored convolutional features combined with transformer-based attention to capture local and global temporal information [4]. For bismuth germanate (BGO) detectors, Loignon-Houle et al. compared LED, a two-threshold time-walk correction and CNN-based timing, showing that performance gains and timing-distribution tails depend on detector characteristics [5].

An open question is whether waveform samples beyond the rising edge improve ML-based LED timing corrections, and which temporal regions contribute most. Previous CNN studies have primarily examined short windows near the rising edge [3, 5], leaving extended-window performance at high sampling rates insufficiently characterized. In related model-based work, Ruiz-Gonzalez et al. observed diminishing returns from additional samples and found that the advantage of maximum-likelihood timing over digital discrimination disappeared above 2 GS/s [6]. These results do not establish whether extending the waveform window benefits data-driven corrections at the 80 GS/s acquisition rate used here.

The analog readout chain is another determinant of timing performance. Amplification can change signal amplitude, bandwidth, noise and rise time, affecting both LED timing and the waveform features accessible to ML. Pourashraf et al. compared front-end readout configurations for silicon photomultiplier (SiPM)-based TOF-PET detectors, reporting a best CTR of 99 ± 2 ps, more than 20 ps better than two alternative configurations [7]. These results establish the importance of electronics choices, but do not determine whether ML corrections applied to outputs of successive stages of cascaded amplification perform differently.

Building on previous waveform-based timing studies, this work investigates how different ML approaches exploit detector signals to correct leading-edge timing, rather than focusing exclusively on achieving the lowest CTR. By comparing multiple estimator families and temporal windows, we aim to characterize the timing information available across the waveform and assess the potential of both conventional and dedicated high-gain readouts.

A further objective is to investigate the scalability of ML-based timing calibration. Training a separate model for every detector pair would become impractical in complex PET systems. We therefore introduce detector-shared estimators, which apply the same correction function to both detector waveforms, enforcing antisymmetry under detector exchange. Comparing these models with unconstrained estimators allows us to assess whether detector-specific relationships are necessary or whether general waveform characteristics are sufficient for effective timing correction.

Although complete waveform acquisition may not be practical in clinical systems, this analysis provides a way to investigate which signal characteristics ML models exploit and which temporal regions are most relevant to their predictions. The two electronics boards are evaluated independently using control, development, and blind acquisitions, with temporal sensitivity analysis complementing the timing-performance evaluation.

2 Experimental setup and datasets

2.1 Detector modules and acquisition

Two opposing detector modules consist of silicon photomultipliers (SiPMs) coupled to $2 \times 2 \times 3\,\mathrm{mm}^3$ lutetium–yttrium oxyorthosilicate (LYSO) crystals, with a ^{22}Na source positioned centrally between them. The University of Chicago (UC) and Fondazione Bruno Kessler (FBK) configurations use the same crystals and SiPMs but different analog front-end electronics.

The two-stage readout chain is shown in Figure 1. Its first stage produces a relatively long signal suitable for energy estimation and conventional timing. The second provides additional amplification and a sharper signal optimized for timing. This dedicated stage requires a separate 5 V supply, adding hardware and integration challenges that make it difficult to deploy in clinical PET systems. The two outputs are referred to as *energy* and *timing*, respectively.

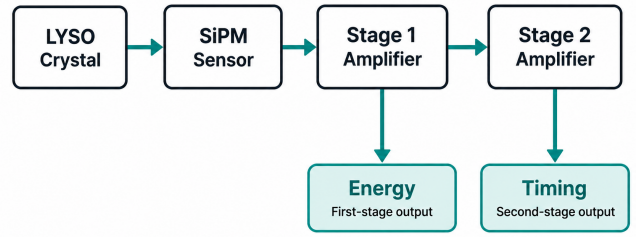


Figure 1: Two-stage readout chain. Energy and timing waveforms are sampled from successive amplification stages of the same SiPM pulse.

The energy signal is acquired without amplitude clipping, and its maximum is used for photopeak selection. For the high-gain timing signal, the oscilloscope's narrower vertical range emphasizes the rising edge while clipping the pulse peak, following a strategy also used by Onishi et al. [3]. In these acquisitions, the energy waveform is cleaner and more stable but exhibits greater time walk; the timing waveform is more susceptible to noise and pulse-shape variations.

2.2 Acquisitions and scientific roles

Each coincidence event contains four simultaneous waveforms: first-stage outputs from detectors 1 and 2 on channels 1 and 2, and second-stage outputs on channels 3 and 4. The 100 ns records are sampled at $80\,\mathrm{GS\,s^{-1}}$ with a SiPM bias of 48 V. The centrally located source defines a true TOF of zero.

For each board, three independent acquisitions are used: *calibration* to determine preprocessing and LED selection criteria, *train* for model training and selection, and *test* for final evaluation. Each calibration dataset contains approximately 20,000 raw coincidence events, while the train and test sets contain approximately 100,000 events each, before event selection. The two boards are analyzed independently.

Figures 2 and 3 show the energy and timing waveforms of the same illustrative coincidence event for UC and FBK, respectively. Both detector signals are shown for each amplification stage. These are raw waveforms before event selection or preprocessing, with only the acquisition vertical-range clipping applied.

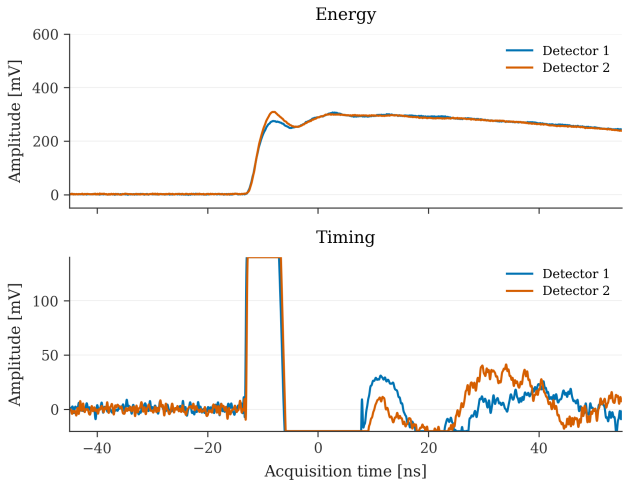


Figure 2: UC raw waveforms: first-stage (energy) and second-stage (timing) signals from both detectors in one event.

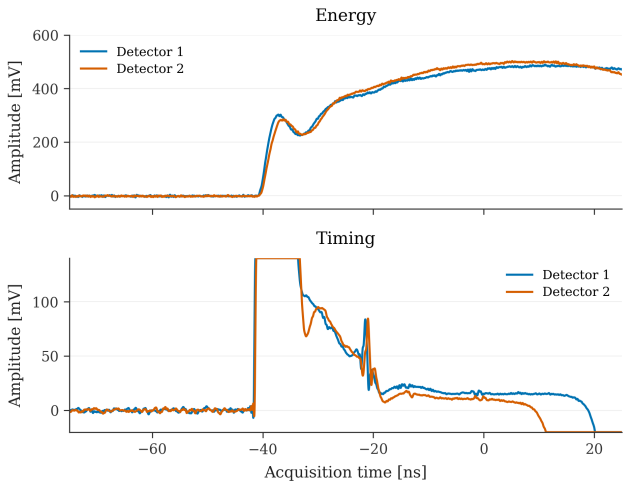


Figure 3: FBK raw waveforms: first-stage (energy) and second-stage (timing) signals from both detectors in one event.

3 Methods

3.1 Event selection and LED calibration

Only photopeak events are retained, based on the maximum amplitudes of both first-stage (energy) waveforms. Acceptance intervals are obtained from iterative Gaussian fits to the photopeaks in the calibration acquisition and applied to both waveform modes. For timing-mode analysis, time-over-threshold (ToT) selection additionally rejects short, spurious crossings associated with noise and signal oscillations. Events with excessive pre-pulse baseline fluctuations or baseline samples near the acquisition limits are excluded. All selection criteria are established from calibration data only.

For each mode, a leading-edge discriminator (LED) threshold is chosen on the calibration acquisition to minimize coincidence timing resolution (CTR), subject to a minimum

fraction of valid coincidences within the prescribed time range. A fixed channel-pair timing offset is subsequently estimated from the calibration LED residuals. Both the threshold and offset remain unchanged in training and testing.

3.2 Timing-error model and learning target

The goal is to correct waveform-dependent errors in a conventional LED measurement, rather than estimate an unconstrained time of flight directly. For a coincidence event, let $\Delta t_{\mathrm{LED}} = t_{1,\mathrm{LED}} - t_{2,\mathrm{LED}}$. We model this measurement as

$$\Delta t_{\mathrm{LED}} = \tau + C + \delta(s_1, s_2) + \epsilon, \tag{1}$$

where τ is the true time of flight and C is the fixed channel-pair offset. The quantities s_1 and s_2 denote the local waveform segments of detectors 1 and 2, respectively. The term $\delta(s_1, s_2)$ represents the waveform-dependent timing error, while ϵ accounts for residual unpredictable variation. The waveform-dependent component δ is the quantity we aim to estimate and correct using ML.

Writing $\widehat{C}$ for the fixed offset estimated during calibration, the supervised target is

$$y = \Delta t_{\mathrm{LED}} - \tau - \widehat{C}, \tag{2}$$

and the corrected time difference is

$$\widehat{\Delta t} = \Delta t_{\mathrm{LED}} - \widehat{C} - h_\theta(s_1, s_2). \tag{3}$$

The final residual relative to true TOF is:

$$r = \widehat{\Delta t} - \tau = y - h_\theta(s_1, s_2). \tag{4}$$

This formulation separates waveform-dependent timing errors, which can potentially be corrected, from unpredictable noise. The source is positioned centrally between the detectors, giving $\tau = 0$. Any positioning error would introduce a constant offset, which can be absorbed into the calibration term without affecting CTR.

Energy and timing waveforms are analyzed separately. Each input waveform is expressed relative to its own LED crossing and normalized using fixed channel-specific amplitude limits. This alignment removes absolute acquisition time from the input. Two LED-relative windows are considered. The short Onishi-inspired window extends from −1.5 to +2 ns and follows the short rising-edge interval used in previous waveform-based timing work [3]. The wide window extends from −2 to +30 ns, covering a substantially larger portion of the available acquisition record. Before constant-sample removal, the two windows contain 279 and 2559 samples per detector, respectively. For the clipped timing signals, constant sample positions are removed using the training population; the resulting input sizes are reported in Table 1. No samples are removed from the energy waveforms. Events lacking a valid coincidence or the full requested window are excluded.

Table 1: Waveform windows relative to the LED crossing and number of retained samples after constant-sample removal.

Board	Signal	Onishi-inspired	Wide
UC	Energy	279	2559
UC	Timing	136	991
FBK	Energy	279	2559
FBK	Timing	138	1944

3.3 Detector-shared and direct corrections

Previous waveform-based timing studies have primarily used direct estimators, which predict the timing correction from the pair of detector waveforms [2, 3]. These models can learn detector-specific relationships without imposing constraints on how the correction changes when the detector inputs are exchanged.

While effective in a two-detector setup, training a separate model for every detector pair would be difficult to scale to a complete PET system. This motivates a detector-shared formulation, in which the correction is expressed as the difference between two evaluations of the same function:

$$h_\theta(s_1, s_2) = g_\theta(s_1) - g_\theta(s_2). \tag{5}$$

The shared function g_θ estimates a waveform-dependent timing contribution for each detector independently. This formulation enforces detector-exchange antisymmetry: swapping the inputs reverses the correction sign, while identical inputs yield zero correction. It also offers a potentially more scalable approach, since the same model can be applied to different physical instances of the same detector type, without requiring parameters specific to each detector pair.

To characterize how the choice of estimator affects waveform-based timing correction, we consider model families with different inductive biases. Linear ridge regression provides a simple and stable baseline. MiniRocket provides an efficient time-series-specific representation. MLPs provide a general nonlinear estimator without explicitly exploiting temporal locality, whereas CNNs incorporate local waveform structure and are included both as a standard time-series architecture and because of their successful use in previous waveform-based timing studies [3, 5]. Locally connected networks retain the local receptive-field structure of CNNs while allowing the learned response to depend on absolute temporal position, which is relevant here because the waveforms are aligned to the LED crossing. Table 2 summarizes the resulting estimators, while implementation details are provided in the appendix.

3.4 Model comparison and training protocol

The independent calibration, training and test acquisitions serve distinct roles, as summarized in Figure 4. Calibration fixes event selection, LED thresholds and timing offsets. Model fitting and selection use only the training set; the test set is reserved for final evaluation.

The models are trained to predict the LED timing error using a standard regression objective, the mean squared error

(MSE):

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left[y_i - h_\theta(s_{1,i}, s_{2,i})\right]^2. \tag{6}$$

Model selection is performed through three-fold cross-validation on the training acquisition, using mean validation CTR as the primary performance criterion. The selected model is then refitted on the full training dataset. The independent test acquisition is subsequently used to compare the observed performance of the tested estimators. References in the Results section to the lowest test CTR therefore identify the best observed result within that specific comparison, rather than a general model ranking. Model-specific training procedures and exceptions are detailed in Appendix A.

3.5 Timing performance and independent evaluation

All configurations are evaluated against LED on the same selected test events. CTR is defined as the full width at half maximum (FWHM) of the timing-residual distribution and computed using histogram-based method F1 of Rainio et al. [8]. Identical histogram settings are used for LED and corrected residuals within each mode. The fixed bin widths are 10 ps for energy and 5 ps for timing.

Root mean squared error (RMSE) relative to true TOF complements CTR because it captures both residual dispersion and systematic bias. In particular, $\mathrm{MSE} = \mathrm{Var}(r) + [\mathbb{E}(r)]^2$; a reduction in RMSE need not imply a comparable narrowing of the central timing peak. Both metrics are therefore needed to characterize the effects of calibration and waveform-based correction.

Fixed preprocessing and the fitted model are applied to the independent test acquisition. Test-event bootstrap resampling provides uncertainties on CTR and RMSE without retraining; these estimates are conditional on the selected fitted model. Within-board comparisons use matched test events and joint resampling to assess paired model differences.

3.6 Temporal sensitivity analysis

To investigate which parts of each waveform influence the correction, temporal occlusion is applied to the trained models on test events. Short groups of samples are replaced by interpolation from neighboring samples, and their importance is quantified by the average absolute change in predicted correction. Shared estimators are perturbed synchronously in both detector inputs, while direct estimators are perturbed one input at a time.

The resulting profiles reveal whether predictions depend primarily on the rising edge or also on later waveform characteristics, and allow comparisons across windows and amplification stages.

Formulation	Code	Estimator	Correction structure
Direct	D-LIN	Linear ridge	Joint linear regression on concatenated waveforms.
	D-MLP	Dense MLP	Joint nonlinear regression on concatenated waveforms.
	D-CNN	Independent CNN	Separate detector-specific CNN scorers; output difference.
	D-ONI	Onishi-inspired CNN	Paired waveforms fused by convolution for joint regression.
	D-MR	MiniRocket + ridge	Multivariate features of the waveform pair; ridge regression.
Shared	S-LIN	Linear ridge	Linear regression on the difference of waveform samples.
	S-MLP	Dense MLP	Common nonlinear detector scorer; output difference.
	S-LCM	Locally connected MLP	Common position-specific local scorer; bounded output difference.
	S-CNN	Temporal CNN	Common convolutional detector scorer; output difference.
	S-MR	MiniRocket + ridge	Common waveform feature transform; feature difference and ridge regression.

Table 2: Waveform-based timing estimators implemented in the benchmark. Model codes are used throughout the results tables. Direct estimators do not enforce detector-exchange antisymmetry; shared estimators use a common detector-level function or feature transform to enforce it.

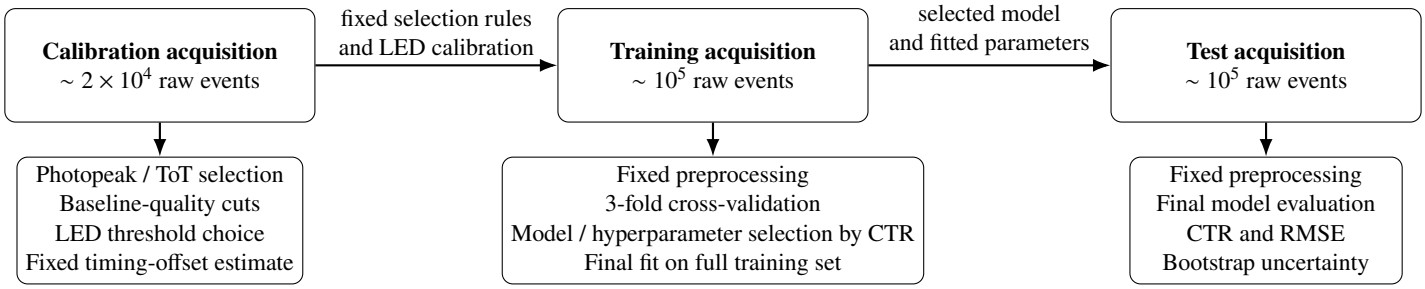


Figure 4: Schematic overview of the experimental protocol. The calibration acquisition is used to determine fixed event-selection and LED-calibration criteria. Model selection and fitting are performed on the independent training acquisition. Final performance is assessed on the independent test acquisition.

4 Results

The aim of the model comparison is not to establish a definitive ranking of estimator architectures. Performance differences can also reflect the specific hyperparameter search and the characteristics of each dataset. The results are therefore interpreted primarily in terms of recurring behavior across different boards, waveform modes, and input windows.

4.1 University of Chicago board

4.1.1 Energy waveforms

Energy-waveform corrections reduce the test CTR from the LED reference of approximately 142 ps to 110.5 ps with the short Onishi-inspired window and 105.3 ps with the wide window (Tables 3 and 4; Figures 10 and 11). The Onishi-inspired paired CNN gives the lowest observed test CTR for the short window, while several shared and direct estimators fall within the same few-picosecond performance range. With the wide window, direct linear ridge gives the lowest observed CTR, again with several other estimators nearby. The relevant trend is the improvement obtained by extending the waveform window rather than the exact ordering of the leading models. Most models show lower CTR and RMSE with the longer window, although the magnitude of the improvement differs by architecture. The independent temporal CNN is a notable exception to the relatively close model agreement in the short-window comparison and remains among the weaker results in the wide-window setting.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	141.9 ± 1.9	142.2 ± 1.7	68.1 ± 1.1
D-ONI	111.0 ± 1.3	110.5 ± 1.2	57.0 ± 1.3
S-MR	111.8 ± 3.3	112.0 ± 1.2	57.4 ± 1.3
S-CNN	113.1 ± 1.6	112.5 ± 1.2	57.6 ± 1.3
D-MR	111.3 ± 3.4	112.5 ± 1.4	57.6 ± 1.3
S-LIN	n/a	112.5 ± 1.2	58.2 ± 1.3
S-LCM	112.9 ± 2.8	112.6 ± 1.2	58.0 ± 1.3
D-LIN	n/a	112.7 ± 1.1	57.6 ± 1.3
S-MLP	112.8 ± 1.4	113.2 ± 1.3	58.4 ± 1.3
D-MLP	112.9 ± 1.0	113.6 ± 1.2	58.0 ± 1.3
D-CNN	112.8 ± 1.3	125.2 ± 1.4	63.1 ± 1.2

Table 3: UC energy onishi: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	141.5 ± 1.7	142.2 ± 1.7	68.0 ± 1.1
D-LIN	n/a	105.3 ± 1.1	54.8 ± 1.4
D-ONI	107.3 ± 2.9	107.3 ± 1.2	55.7 ± 1.4
S-LIN	n/a	107.4 ± 1.3	55.4 ± 1.3
S-LCM	107.8 ± 1.0	107.5 ± 1.3	55.5 ± 1.3
S-CNN	105.2 ± 0.5	108.0 ± 1.2	55.6 ± 1.3
S-MLP	106.3 ± 1.9	108.2 ± 1.3	55.6 ± 1.4
D-MR	108.5 ± 1.9	109.0 ± 1.2	56.5 ± 1.3
S-MR	109.8 ± 0.3	109.0 ± 1.4	56.4 ± 1.4
D-MLP	131.0 ± 3.9	115.6 ± 1.3	58.7 ± 1.3
D-CNN	128.0 ± 9.3	123.5 ± 1.6	62.4 ± 1.2

Table 4: UC energy wide: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	57.6 ± 0.6	57.1 ± 0.7	36.6 ± 1.7
S-MLP	57.3 ± 1.5	56.3 ± 0.7	36.4 ± 1.7
S-LIN	n/a	56.4 ± 0.6	36.3 ± 1.6
D-ONI	56.9 ± 0.3	56.5 ± 0.7	36.7 ± 1.7
S-LCM	57.0 ± 0.9	56.6 ± 0.7	36.4 ± 1.7
D-LIN	n/a	56.7 ± 0.6	36.2 ± 1.7
D-CNN	57.6 ± 0.4	56.7 ± 0.7	36.6 ± 1.7
S-CNN	57.5 ± 1.1	56.8 ± 0.6	36.3 ± 1.7
D-MLP	57.5 ± 0.4	56.9 ± 0.7	36.6 ± 1.7
S-MR	57.1 ± 0.3	57.3 ± 0.6	35.3 ± 1.2
D-MR	68.8 ± 2.8	64.9 ± 0.7	38.0 ± 1.2

Table 5: UC timing onishi: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

4.1.2 Timing waveforms

The timing-waveform LED reference is approximately 57.1 ps. With the short window, most estimators remain close to this reference, and the lowest observed test CTR is 56.3 ps. With the wide window, the minimum decreases to 52.9 ps, and several linear, MiniRocket-based and shared estimators lie between approximately 53 and 54 ps (Tables 5 and 6; Figures 12 and 13). Thus, extending the timing input produces a clearer improvement in CTR than the small short-window corrections. The short-window direct MiniRocket result is an outlier, with a test CTR exceeding the LED reference. RMSE changes are generally smaller than the corresponding CTR changes.

4.1.3 Temporal sensitivity: wide window

The temporal-occlusion maps for the lowest-test-CTR wide-window estimators are shown in Figures 5 and 6. For energy waveforms, the largest sensitivity is concentrated around LED crossing and the early rising edge, with weaker sensitivity over the later waveform. In timing mode, sensitivity is also concentrated near the crossing and the end of the clipped portion, while a smaller response appears around the late feature at approximately 21–22 ns. The latter is visible in both detector traces, but its importance is much lower than that of the early regions.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	57.9 ± 0.9	57.1 ± 0.7	36.6 ± 1.7
D-MR	54.1 ± 0.3	52.9 ± 0.8	35.6 ± 1.7
S-MR	54.9 ± 1.3	53.5 ± 0.8	35.4 ± 1.6
D-LIN	n/a	53.5 ± 0.7	35.4 ± 1.7
S-LIN	n/a	53.7 ± 0.7	35.5 ± 1.7
S-CNN	54.8 ± 0.8	54.5 ± 0.6	35.6 ± 1.7
D-MLP	55.1 ± 1.3	54.9 ± 0.6	35.9 ± 1.8
D-ONI	55.0 ± 0.4	55.0 ± 0.6	35.8 ± 1.7
S-LCM	54.3 ± 0.6	55.4 ± 0.5	36.0 ± 1.7
S-MLP	54.9 ± 1.2	55.8 ± 0.6	36.0 ± 1.7
D-CNN	57.6 ± 1.6	56.9 ± 0.7	36.6 ± 1.7

Table 6: UC timing wide: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

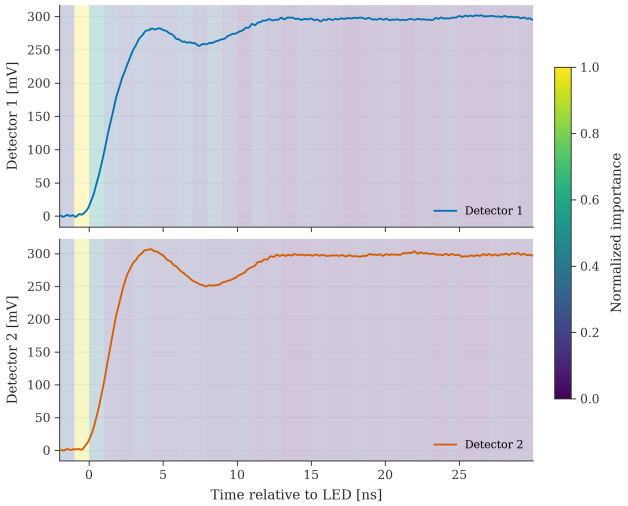


Figure 5: UC energy waveforms, wide window: temporal occlusion sensitivity of the estimator with the lowest observed test CTR (direct linear ridge), shown against the two detector waveforms. Colors indicate normalized sensitivity to local occlusion, not causal attribution.

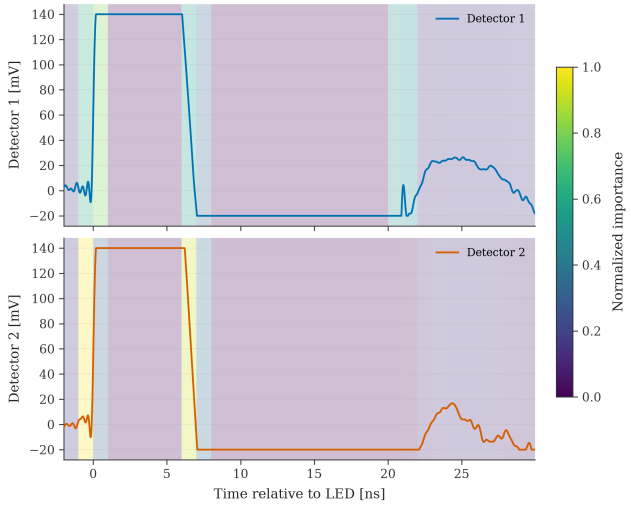


Figure 6: UC timing waveforms, wide window: temporal occlusion sensitivity of the estimator with the lowest observed test CTR (direct MiniRocket), shown against the two detector waveforms.

4.2 Fondazione Bruno Kessler board

4.2.1 Energy waveforms

For FBK energy waveforms, the LED reference is approximately 147.3 ps. The lowest observed short-window CTR is 115.4 ps, with linear and nonlinear estimators occupying a similar performance range (Table 7; Figure 14). The wide-window minimum is 111.0 ps, obtained with direct MiniRocket, while several other estimators remain around 114–116 ps (Table 8; Figure 15). The shared temporal CNN is a marked wide-window outlier: despite a validation CTR of approximately 115 ps, the final refit provides essentially no improvement over LED on the independent test set. Neural-network optimization is non-convex and stochastic, so different weight initializations, minibatch sequences, or changes in the fitting population can lead the same architecture and hyperparameters to different fitted solutions. The observed discrepancy is therefore consistent with an unstable final optimization, although the present analysis does not isolate its specific cause. As with UC, the strongest wide-window results improve over the short-window minimum, but individual architectures do not all benefit equally.

4.2.2 Timing waveforms

In FBK timing mode, the LED reference is about 62.0 ps. Short-window test CTRs range from approximately 57.7 ps to values close to LED, whereas the wide-window comparison reaches a minimum of 54.4 ps (Tables 9 and 10; Figures 16 and 17). The leading wide-window results are obtained with linear ridge and MiniRocket-based estimators; several shared models are also competitive. Conversely, some neural estimators provide little additional narrowing of the timing distribution in this configuration. The wider window produces a lower best CTR and a modest reduction in best RMSE.

4.2.3 Temporal sensitivity: wide window

The FBK energy map (Figure 7) shows its greatest sensitivity near LED crossing, with lower and more diffuse sensitivity in the following waveform. The timing map (Figure 8)

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	145 ± 3	147.3 ± 1.8	70.9 ± 1.0
D-ONI	114.9 ± 1.5	115.4 ± 1.2	60.6 ± 1.2
S-LIN	n/a	115.9 ± 1.3	62.8 ± 1.2
D-LIN	n/a	116.2 ± 1.5	61.4 ± 1.2
D-MR	115.9 ± 2.3	116.4 ± 1.2	60.8 ± 1.2
S-LCM	117.3 ± 2.4	116.5 ± 1.4	62.0 ± 1.0
S-CNN	116.9 ± 0.1	116.7 ± 1.4	61.8 ± 1.1
D-MLP	118.6 ± 1.7	117.7 ± 1.2	61.5 ± 1.1
D-CNN	123.9 ± 4.5	117.7 ± 1.2	61.2 ± 1.2
S-MLP	116.9 ± 1.9	119.0 ± 1.4	62.8 ± 1.2
S-MR	116.8 ± 0.8	120.4 ± 1.1	61.5 ± 1.1

Table 7: FBK energy onishi: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	145 ± 1	147.3 ± 1.6	70.9 ± 1.1
D-MR	111.9 ± 0.2	111.0 ± 1.2	58.6 ± 1.0
D-LIN	n/a	114.4 ± 1.1	59.9 ± 1.3
S-LCM	114.7 ± 0.7	114.9 ± 1.2	60.1 ± 1.3
D-ONI	114.1 ± 0.9	115.3 ± 1.4	59.4 ± 1.1
S-LIN	n/a	115.9 ± 1.2	60.2 ± 1.3
S-MLP	115.3 ± 0.8	115.9 ± 1.4	60.3 ± 1.2
D-MLP	119.6 ± 2.4	116.9 ± 1.3	60.9 ± 1.1
S-MR	117.0 ± 1.6	117.9 ± 1.4	59.6 ± 1.1
D-CNN	131.6 ± 3.4	130.6 ± 1.2	65.2 ± 1.1
S-CNN	115.0 ± 0.2	147.3 ± 1.7	70.9 ± 1.1

Table 8: FBK energy wide: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	62.0 ± 0.3	62.0 ± 0.8	42.3 ± 1.7
S-CNN	57.7 ± 2.0	57.7 ± 0.8	41.4 ± 1.7
S-MR	56.8 ± 0.7	58.0 ± 0.7	41.3 ± 1.8
D-LIN	n/a	58.1 ± 0.6	41.4 ± 1.5
S-LIN	n/a	58.2 ± 0.6	41.4 ± 1.9
D-ONI	56.9 ± 1.5	58.3 ± 0.5	41.4 ± 1.7
S-MLP	58.1 ± 1.8	58.8 ± 0.6	41.6 ± 1.8
D-MR	57.4 ± 2.1	59.1 ± 0.6	41.3 ± 1.7
S-LCM	57.9 ± 1.4	59.5 ± 0.6	41.7 ± 1.8
D-MLP	60.6 ± 1.9	61.8 ± 0.7	42.3 ± 1.7
D-CNN	61.5 ± 0.5	62.0 ± 0.8	42.3 ± 1.6

Table 9: FBK timing onishi: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

Model	CV CTR [ps]	Test CTR [ps]	Test RMSE [ps]
LED	62.2 ± 1.3	62.0 ± 0.8	42.3 ± 1.7
D-LIN	n/a	54.4 ± 0.7	40.3 ± 1.6
D-MR	54.4 ± 0.7	55.1 ± 0.6	40.2 ± 1.7
S-MR	55.2 ± 1.8	55.9 ± 0.6	40.4 ± 1.8
S-LIN	n/a	55.9 ± 0.5	40.4 ± 1.9
S-MLP	55.8 ± 1.1	56.3 ± 0.6	40.9 ± 1.8
S-CNN	55.8 ± 0.2	57.5 ± 0.7	40.8 ± 1.7
S-LCM	56.4 ± 1.4	57.6 ± 0.6	41.1 ± 1.9
D-ONI	58.1 ± 3.5	61.8 ± 0.8	42.3 ± 1.7
D-CNN	62.1 ± 1.3	61.9 ± 0.8	42.3 ± 1.6
D-MLP	61.9 ± 1.3	62.0 ± 0.7	42.3 ± 1.7

Table 10: FBK timing wide: validation CTR (mean ± fold SD), independent-test CTR and RMSE (estimate ± bootstrap SD). Models are ranked by unrounded test CTR; codes are defined in Table 2. LED has no outer CV; ridge uses internal LOOCV.

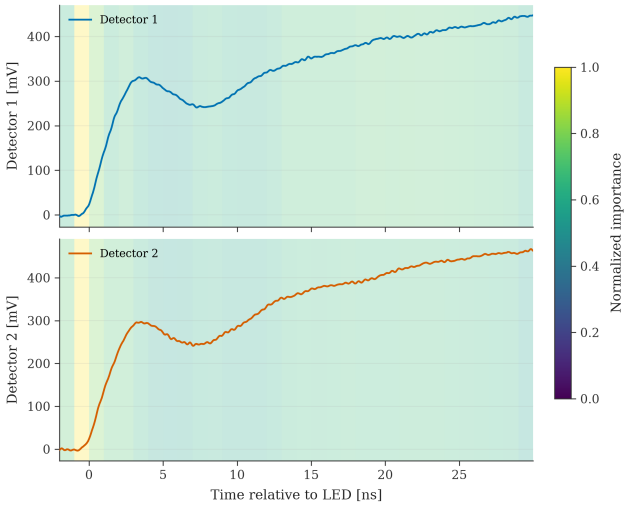


Figure 7: FBK energy waveforms, wide window: temporal occlusion sensitivity of the estimator with the lowest observed test CTR (direct MiniRocket), shown against the two detector waveforms.

likewise emphasizes the crossing, but also shows a secondary sensitive region around 20 ns, near a sharp feature in the later signal. The additional shared MiniRocket map in Figure 9 makes this contrast especially clear: although it is not the lowest-CTR estimator, its sensitivity contains a prominent secondary maximum around 20–21 ns, as well as a smaller feature near 8–9 ns. These profiles quantify changes in model predictions under the specified occlusion perturbation; they do not separately establish the physical origin or independent information content of the highlighted regions.

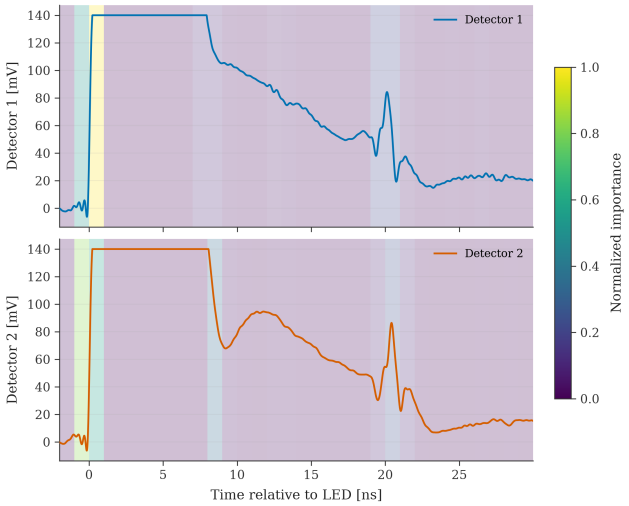


Figure 8: FBK timing waveforms, wide window: temporal occlusion sensitivity of the estimator with the lowest observed test CTR (direct linear ridge), shown against the two detector waveforms.

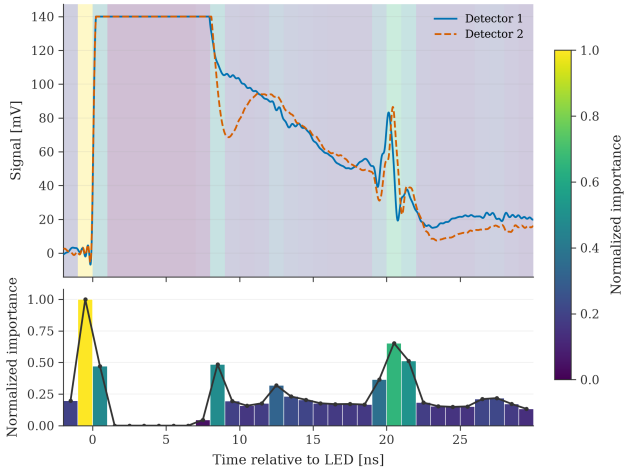


Figure 9: FBK timing waveforms, wide window: shared MiniRocket temporal occlusion sensitivity, included as an additional model comparison rather than the lowest-observed-CTR estimator. The lower panel displays the normalized sensitivity profile, including its secondary peak near 20–21 ns.

4.3 Comparison across waveform modes

Across both boards, the dedicated second-stage timing signals provide substantially better timing resolution than the first-stage energy signals, both before and after waveform-based correction. In the wide-window comparisons, the minimum energy CTRs are 105.3 and 111.0 ps for UC and FBK, respectively, compared with 52.9 and 54.4 ps for timing waveforms (Tables 4, 6, 8, and 10). Importantly, waveform-based correction still provides an additional improvement when applied to the timing-optimized second-stage signal. The two boards reach a similar corrected timing performance despite their different LED references and front-end implementations.

5 Conclusions

5.1 CTR improvement

Waveform-based correction substantially improved leading-edge timing on the first-stage energy signals of both boards. The test CTR was reduced by more than 20% even with the short Onishi-inspired window, with reductions reaching approximately 25% when the wider window was used. The strong performance of the Onishi-inspired CNN in the short-window energy comparisons is consistent with previous waveform-based LED correction studies [3, 5].

5.2 Shared detector-level models

The detector-shared formulation retained essentially the same useful timing correction as the direct formulation without using separate detector-specific scoring functions. Across boards, waveform modes, and windows, shared estimators generally remained in the same performance range as the leading direct estimators. The small few-picosecond differences between the best shared and direct results are not interpreted as evidence that either formulation is intrinsically superior, since such differences may depend on the detector and acquisition setup. The main result is that most of the achievable waveform-based correction can be retained with a shared detector-level model.

5.3 Model comparison

CNN and MLP approaches did not provide a systematic advantage over simpler estimators, despite the successful application of CNNs in previous waveform-based timing studies [3, 5]. Across the tested configurations, linear ridge regression and MiniRocket features followed by ridge regression reached the same overall performance range as the neural models. Because the leading estimators often differ by only a few picoseconds, their exact ordering is not considered a general property of the model families and may change with detector and acquisition conditions. The broader result is that much of the waveform-dependent timing correction can be captured without requiring end-to-end nonlinear neural networks. The marked FBK energy-wide shared-CNN failure also highlights a practical advantage of the ridge-based estimators: for fixed input features, ridge regression has a convex objective and is not subject to the initialization-dependent optimization of the neural models. Together with their competitive timing performance, this makes the ridge-based approaches particularly attractive for this correction task.

5.4 Window length and XAI

Extending the waveform window consistently improved the achievable timing resolution on both boards, with a particularly important effect on the timing signals, where short windows provided limited correction. Temporal occlusion analysis showed that energy-channel models relied primarily on the region around the LED crossing, while timing-channel models were also sensitive to later waveform structures. This was especially evident for the FBK board, where a pronounced secondary sensitivity peak appeared around 20 ns. These findings support the usefulness of waveform information beyond the rising edge.

5.5 Final considerations

Despite ML correction, the second-stage timing signals achieved approximately twice the timing resolution of the first-stage energy signals, highlighting that waveform-based correction cannot fully compensate for differences in analog readout quality. Nevertheless, ML consistently improved LED timing, with both boards reaching similar corrected CTRs of approximately 53–54 ps despite different amplification electronics and LED thresholds. This convergence suggests that the remaining timing uncertainty may be dominated by physical effects common to the detectors that cannot be recovered from the available waveform information, although this hypothesis requires further investigation.

A Model architectures and training settings

The following model families represent different assumptions about how waveform information can correct LED timing. Their scientific motivation and detector-pair formulations are described before the configured hyperparameter tables.

A.1 Linear ridge regression

Linear models provide a computationally inexpensive baseline for testing whether a weighted combination of waveform samples is sufficient for timing correction. The direct estimator uses independent weights on the concatenated detector waveforms and an intercept. The shared estimator instead applies common weights to the sample-wise waveform difference, without an intercept, enforcing detector-exchange antisymmetry. By linearity, the shared correction can be written as

$$h(s_1, s_2) = \mathbf{w}^T(s_1 - s_2) = \mathbf{w}^T s_1 - \mathbf{w}^T s_2. \tag{7}$$

Thus, although the model is trained on waveform differences, the same fitted weights can be applied independently to each detector waveform, with only the resulting scalar scores subtracted. Ridge regularization limits sensitivity to highly correlated samples. Its strength is selected by internal leave-one-out cross-validation minimizing mean squared error (MSE), rather than the outer CTR-based selection.

A.2 Convolutional neural networks

Convolutional neural networks (CNNs) are established waveform-timing models, including for LED correction [3, 5]. Their temporal filters exploit local waveform structure with weights shared across sample positions. The Onishi-inspired direct CNN combines the detector dimension in the first convolution and predicts a joint correction. The independent CNN uses separately parameterized temporal backbones and scalar heads for the two detector signals, then subtracts the outputs. The shared CNN uses one backbone and head for both signals, imposing the antisymmetric correction. Comparing these two variants tests the effect of detector-level parameter sharing.

A.3 Multilayer perceptrons

Because the waveforms are aligned to their individual LED crossings, relevant features occur at approximately consistent input positions, so translation-invariant representations are not necessarily required. A direct dense multilayer perceptron (MLP) regresses on the concatenated detector waveforms. The shared MLP applies a single dense network g_θ to each detector waveform independently, using identical learned parameters. The correction is obtained by subtracting the two scalar outputs:

$$h_\theta(s_1, s_2) = g_\theta(s_1) - g_\theta(s_2). \tag{8}$$

The network is trained end-to-end on waveform pairs, minimizing the prediction error of the resulting correction against the supervised timing target. Both branches contribute to updating the same parameters, thereby enforcing detector-exchange antisymmetry throughout training and inference.

The shared locally connected MLP preserves temporal neighborhoods without sharing convolutional filters across positions. Two successive layers have independent kernels at each temporal position, followed by a dense detector scorer. An odd, smoothly bounded transformation is applied to the score difference, limiting correction amplitude while preserving antisymmetry. Local connectivity provides an intermediate architectural assumption between dense and convolutional processing [9].

A.4 MiniRocket feature regression

MiniRocket is an efficient, largely deterministic convolutional feature transformation developed for time-series classification [10]. We use its features for timing-error regression with ridge regularization. The direct model applies a multivariate MiniRocket transform jointly to the two signals; the shared model applies a common univariate transform to each detector and regresses on their feature difference, without an intercept. Feature transforms and scaling are fitted on the calibration acquisition and fixed before supervised fitting. Unlike the two linear ridge estimators, MiniRocket models select ridge regularization using outer CTR-based cross-validation.

A.5 Neural training procedures

Dense MLPs, the locally connected MLP and temporal CNNs minimize root mean squared error (RMSE) using an internal training holdout for early stopping. The shared dense MLP uses Nesterov stochastic gradient descent; the other models in this group use Adam. The Onishi-inspired paired CNN instead minimizes MSE using Adam and a fixed epoch schedule without an internal early-stopping holdout. For all these neural models, the hyperparameters are selected using mean held-out coincidence timing resolution (CTR) in the three common outer folds, even though the fitting loss is squared-error-based.

A.6 Configured hyperparameter spaces

Tables 11 and 12 list the dense, locally connected and temporal CNN settings. Table 13 describes the paired CNN, while Table 14 gives the linear and MiniRocket settings. Square brackets indicate sequences of layer widths or channels, braces indicate alternative values, and stated ranges with a count use logarithmic spacing.

Formulation	Estimator	Parameter	Configured space or value
Shared	Dense MLP	Dense hidden widths	{[16], [128], [128, 64], [16, 16, 16]}
		Activation	ReLU
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	32
		Optimizer	Nesterov SGD; momentum 0.9
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
	Locally connected MLP	Dense hidden widths	[128]
		Activation	ReLU
		First local kernel size	3
		First local stride	1
		Second local kernel size	3
		Second local stride	1
		Smooth correction bound [ps]	500
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
Direct	Dense MLP	Dense hidden widths	{[16], [128], [128, 64], [16, 16, 16]}
		Activation	ReLU
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	32
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4

Table 11: Dense and locally connected neural model spaces. Neural losses use RMSE. Gradient-based stopping is enabled with the listed threshold and patience.

Formulation	Estimator	Parameter	Configured space or value
Shared	Temporal CNN	Dense hidden widths	[64]
		Activation	ReLU
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Convolutional channels	[16, 32]
		Temporal kernel sizes	[5, 3]
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
Direct	Independent temporal CNN	Dense hidden widths	[64]
		Activation	ReLU
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Convolutional channels	[16, 32]
		Temporal kernel sizes	[5, 3]
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4

Table 12: Temporal CNN model spaces. Kernels are valid convolutions without pooling; the dense head follows flattened convolutional features. Neural losses use RMSE.

Formulation	Estimator	Parameter	Configured space or value
Direct	Onishi-inspired paired CNN	Convolutional channels	[32, 64]
		Temporal kernel sizes	[5, 3]
		Dense hidden width	256
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Optimizer	Adam
		Maximum epochs	100
		Learning-rate milestones [epochs]	[30, 60]
		Learning-rate decay factor	0.1

Table 13: Paired CNN settings. The first kernel spans both detectors; later kernels span one detector-height position. Training uses MSE on the complete fitting subset, without an internal early-stopping holdout.

Formulation	Estimator	Parameter	Configured space or value
Shared	Linear ridge	Regularization grid	50 logarithmically spaced values from 10^{-8} to 1000
		Internal selection	Leave-one-out CV; MSE
		Intercept	None
	MiniRocket	Requested MiniRocket features	10000
		Ridge regularization strengths	{0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000, 10000, 30000, 100000, 300000, 1e+06}
		Feature fitting	Control acquisition; scaling without centring
		Regression intercept	None
		Selection	Outer CV; mean CTR
Direct	Linear ridge	Regularization grid	50 logarithmically spaced values from 10^{-8} to 1000
		Internal selection	Leave-one-out CV; MSE
		Intercept	Included
	MiniRocket	Requested MiniRocket features	10000
		Ridge regularization strengths	{0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000, 10000, 30000, 100000, 300000, 1e+06}
		Feature fitting	Control acquisition; scaling without centring
		Regression intercept	Included
		Selection	Outer CV; mean CTR

Table 14: Linear and MiniRocket spaces. Only the two linear estimators use internal ridge CV. MiniRocket transforms are frozen before outer supervised validation.

A.7 Software libraries

Table 15 summarizes the scientific software used. Version bounds are declared minimum requirements rather than measured installation versions.

Library	Declared minimum	Role
PyTorch	2.2	Dense, locally connected and convolutional neural networks; tensor differentiation and training.
scikit-learn	1.4	Ridge regression, internal ridge CV and feature scaling.
sktime	0.35	Univariate and multivariate MiniRocket transforms.
NumPy	1.26	Waveform arrays, timing operations and bootstrap sampling.
SciPy	1.11	Scientific numerical routines and Gaussian fits for event selection.
Matplotlib	3.8	Timing, comparison and temporal-sensitivity figures.
pandas	2.1	Tabular data and transformed features.
Optuna	4.0	Optional adaptive parameter proposal; supplied benchmark spaces use finite grids.

Table 15: Model libraries and supporting scientific software.

B Independent-test CTR comparisons

The following figures present the model-wise independent-test CTR for the two waveform modes and input windows on each board. Corresponding numerical results and uncertainty estimates are provided in Section 4.

B.1 UC board

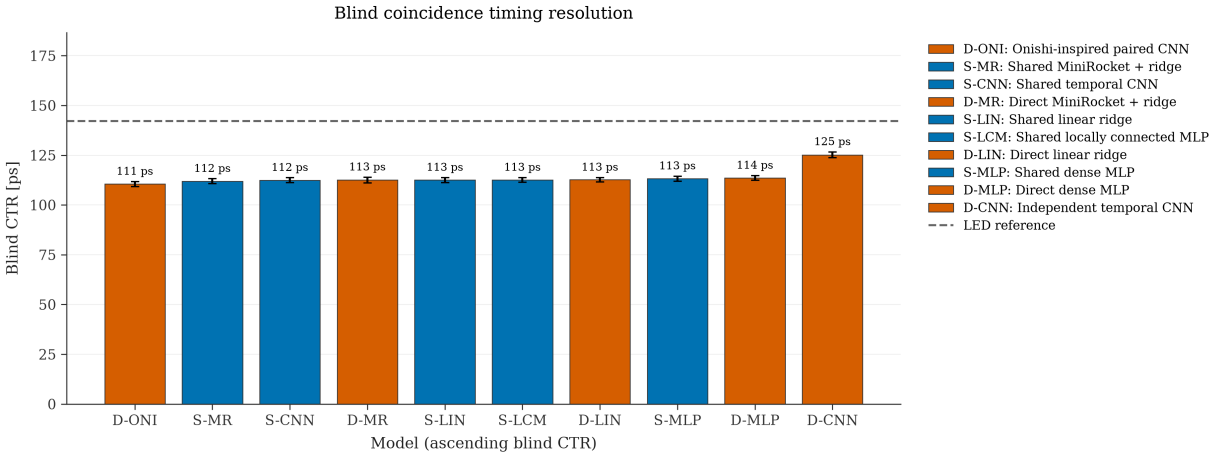


Figure 10: UC board, energy waveforms, short Onishi-inspired window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

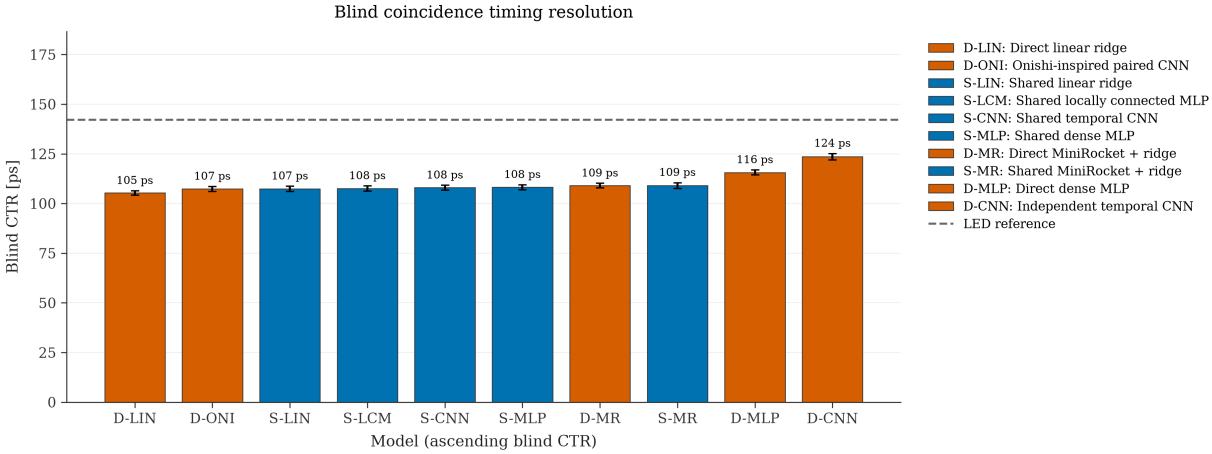


Figure 11: UC board, energy waveforms, wide window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

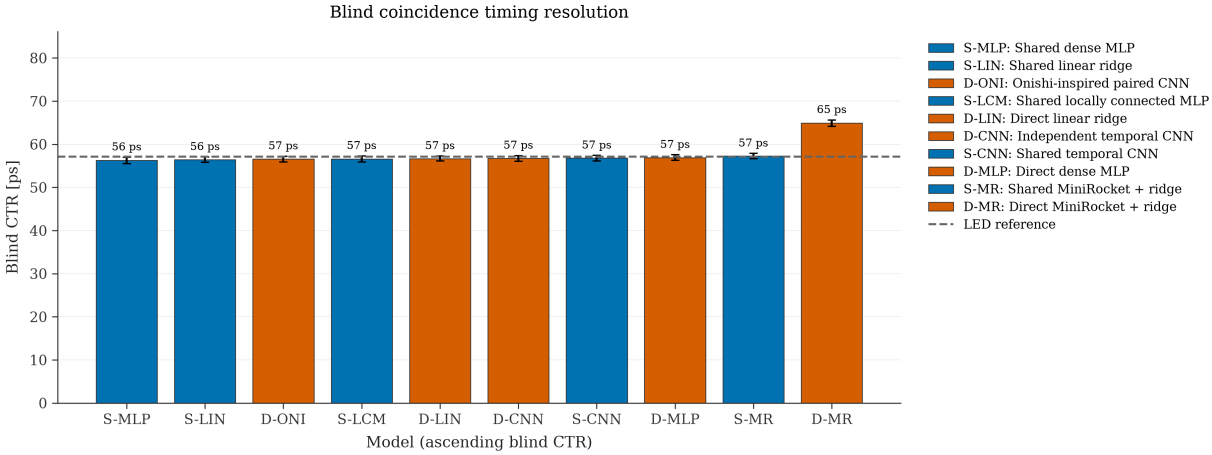


Figure 12: UC board, timing waveforms, short Onishi-inspired window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

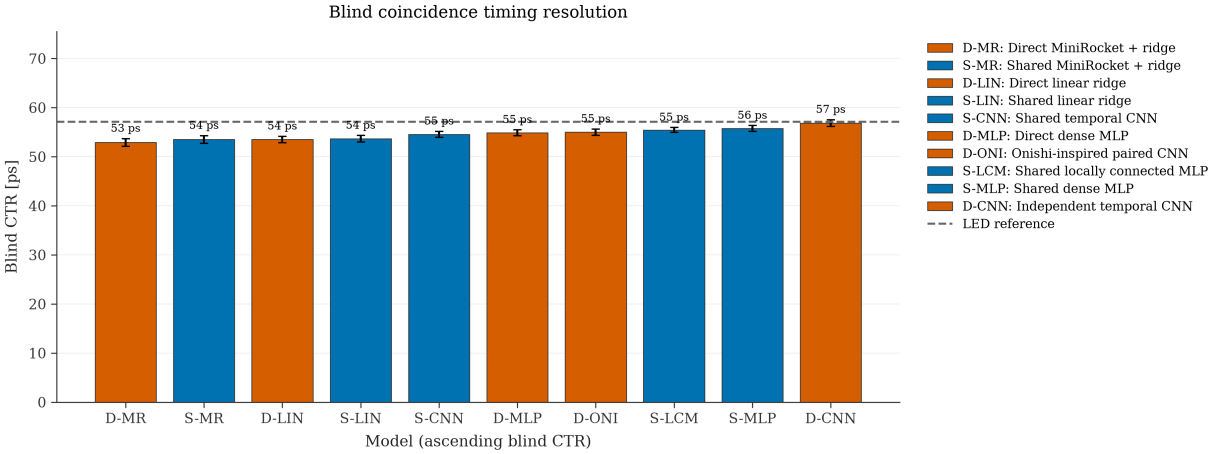


Figure 13: UC board, timing waveforms, wide window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

B.2 FBK board

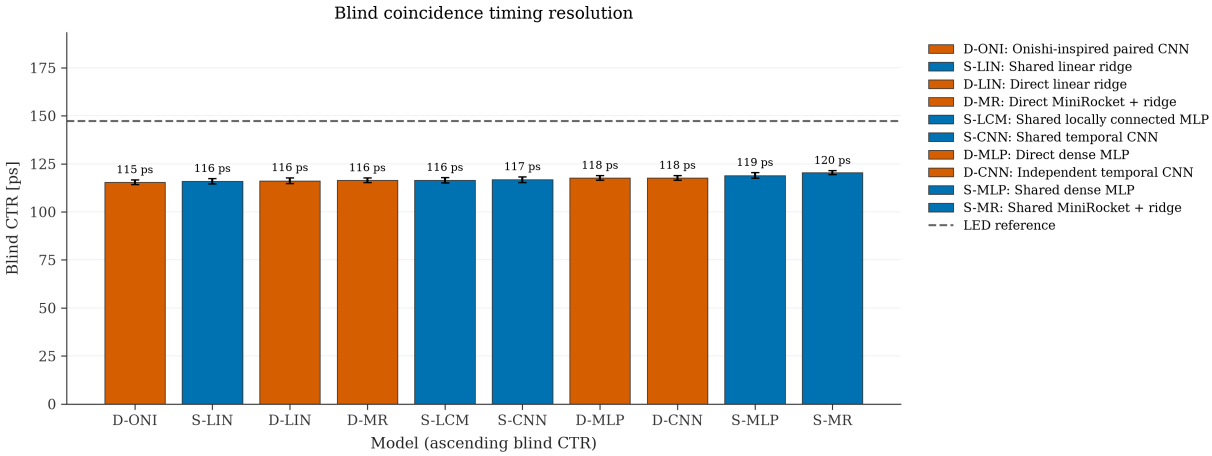


Figure 14: FBK board, energy waveforms, short Onishi-inspired window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

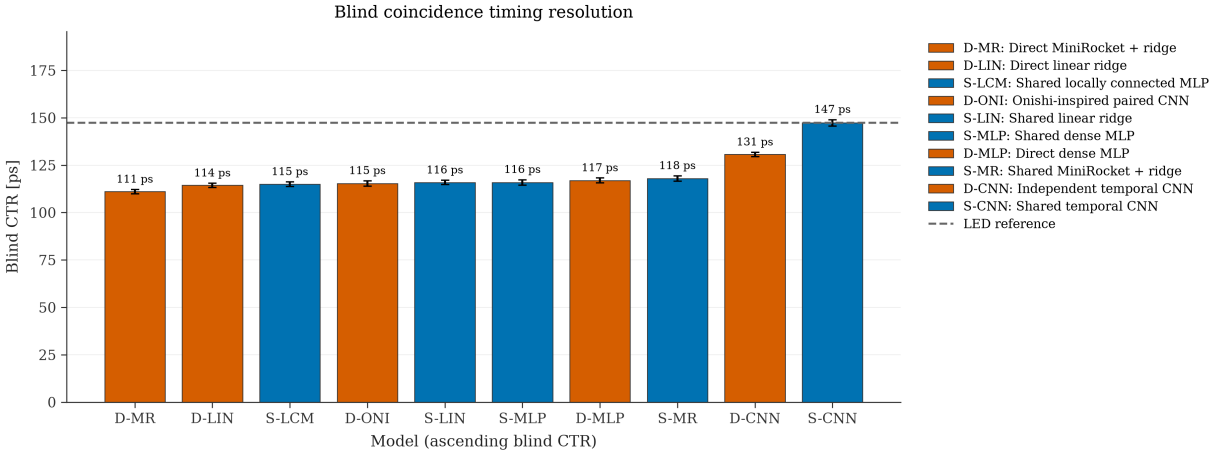


Figure 15: FBK board, energy waveforms, wide window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

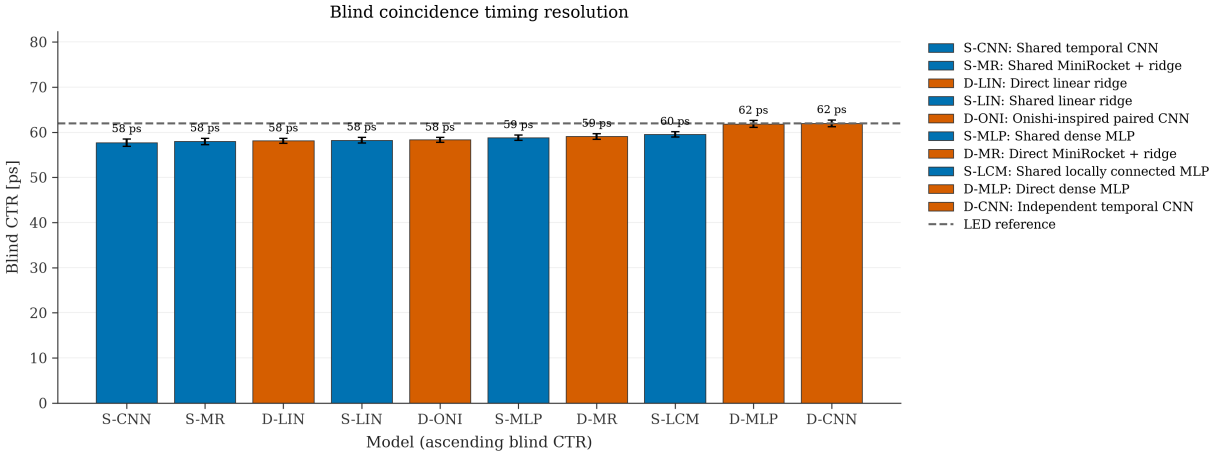


Figure 16: FBK board, timing waveforms, short Onishi-inspired window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

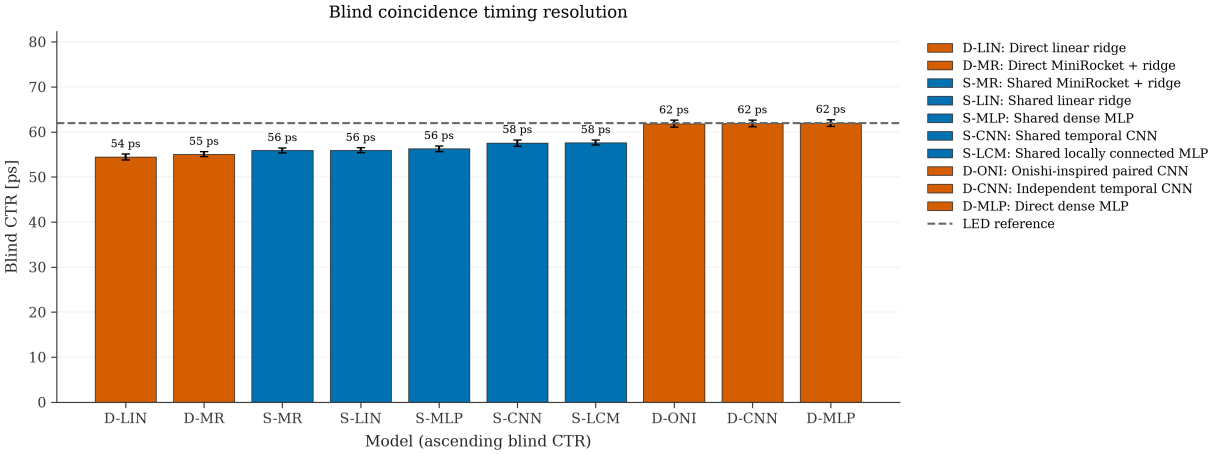


Figure 17: FBK board, timing waveforms, wide window: independent-test CTR of the compared estimators. The dashed line denotes the LED reference; error bars indicate bootstrap standard deviations.

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